

THE STARSHIP GALILEO

DEEP SPACE TRAVEL AND COLONIZATION

INSTRUCTIONS MANUAL

THE GEMINI PROTOCOLS



PHASE 03

THE BOOTSTRAP LAUNCH STORES

Initial onboard helium for the plasma — water for the hydrogen — Nitropril. Carried until speed is at cruise collection speed. With the EM collection array, the funnel is a field, not a structure

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THE GEMINI PROTOCOLS — CHAPTER 03

PHASE 03: THE BOOTSTRAP LAUNCH STORES — INITIAL ONBOARD HELIUM FOR THE PLASMA, WATER FOR THE HYDROGEN, NITROPRIL — with the EM Collection Array. Status: CURRENT. Renamed 12 August 2026 (naming batches). Chapter GP-IM-003 v1.0 — 19 August 2026.

§0 — READ THIS FIRST (HOW TO USE THIS CHAPTER)

Written for idiot-proofing — for AIs and for speed. (1) Build exactly what is written. (2) Cross-phase directions are INLINED at §7 — never a pointer. (3) Rejected options and honest limits are §8. (4) Owed-to-bench items are §9. (5) Change control §10 — only the Author's orders change the design; the master always wins.

STANDARD NOTATION — MATERIALS & CALIBRATIONS (series law, sitting 461): Materials or calibrations is not missing science or engineering — it is, to any real physicist or engineer, stating the flkn obvious. Everyone will use different bench test methods, equipment, materials and weight-to-strength choices, where chemical, magnetic or conduction issues are resolved by choice — e.g. carbon fibre over steel. Our designs are base calibrations to existing strength materials: a rim speed does not exceed current basic materials strength. We do not spec methods, and we do not teach welders how to weld.

§1 — WHAT YOU ARE BUILDING

Two systems in one phase. FIRST: the bootstrap stores — the consumables every ship carries to bridge from departure to the moment the collectors come online at the gas giants — there is no cruise collection speed and no threshold; collection happens at the gas giants, at orbital speeds (the phase 01 law). SECOND: the EM collection array — the field funnel that refills the stores at the gas giants. The stores carry the ship to cruise; the array is what refills them at cruise.

§2 — THE DOCTRINE (THE SIZING LAW)

- Every ship carries its bootstrap: consumables sized to bridge departure -> collection speed.
- THE SIZING LAW: consumption rate × ramp time to cruise collection speed, PLUS MARGIN.
- THE STORE IS A BRIDGE, NOT A CRUTCH: the collectors must actually take over at cruise — or the doctrine fails honestly and the mass budget says so on day one.
- NO NEW PHYSICS REQUIRED: the oldest sound pattern in exploration — carry what you need until the environment provides it (ISRU staging, ram-collection doctrine). Onboard stores need mass budget and honest consumption numbers, nothing else.

§3 — THE STORES (WHAT GOES ABOARD)

- HELIUM — the plasma cores are HELIUM machines, backup only, no longer in the engine chain; their working fluid comes from this store, and the harvest refills it at the gas giants. The ship climbs on the microwave-hydrogen main engine — the cores do not turn for the climb.
- WATER — for the hydrogen line: electrolysis on demand, the file's own un-fckupable closed-loop plumbing (factory economy inlined §7.3).
- NITROPRIIL — the atmospheric stabilizer family: solid-state, un-pressurized granules (Phase 88), synthesizable aboard from raw atmospheric nitrogen (Phase 128) — directions inlined §7.4. No high-risk gaseous or liquid nitrogen storage.

§4 — THE EM COLLECTION ARRAY (AMENDMENT 1 — THE SKINNY)

THE TRICK — THE FUNNEL IS A FIELD, NOT A STRUCTURE. A physical scoop can only be as wide as what you build. An EM array cheats: superconductor loops project a magnetic field kilometres wide, and charged particles that cross it spiral down the field lines into a throat that is metres wide. A 10 m loop wears a 2 km invisible net. Effective collection area = the field's cross-section, not the ship's. Proven physics running free tonight: Earth's magnetosphere funnels solar wind into the poles — an aurora is a planet running a collection array.

- CATCH ONE — MAGNETS ONLY GRAB CHARGE. Raw interstellar hydrogen is mostly neutral; a neutral atom does not know your field exists. A real array needs an IONIZER: a forward laser or electron beam that strips the gas before it arrives, far enough ahead that the fresh ions have room to curve into the throat. That is a power bill, and it is why most paper ramjets die. The hardware family is already owned (§7.2).
- CATCH TWO — THE RAM TAX. Everything you catch arrives at your speed and must be re-thrown. Net momentum per kilo = exhaust speed minus ship speed. At 30,000 mph (13.4 km/s) against the 50 km/s helium exhaust, catch-and-throw is NET THRUST — the cores pay the energy, the medium supplies the mass, the tanks refill. Cross 50 km/s ship speed and the same array becomes a brake. 'Cruise collection speed' has an honest definition: BELOW EXHAUST SPEED.
- CATCH THREE — THE MIRROR BITES. A converging field reflects a share of incoming ions straight back out (pitch-angle mirroring). You never collect everything that enters the net; throat geometry decides the fraction. Design knob, not a killer.
- CATCH FOUR — DENSITY IS STILL THE TYRANT. At cruise through a 2 km funnel in raw ISM (~ 1 atom/cm³) the array collects a few grams per year. The array does not beat the wall; it never did. The array is the NET — the seeded lane (highway buoys, 242 mini-moons) is the STOCKED RIVER. Net without river is jewellery.

- BONUS REGIME: inside a star system the solar wind arrives PRE-IONIZED — the laser gets a holiday and the scoop eats free on departure and arrival.
- THE VERDICT (seated): real physics, right side of the momentum ledger at cruise speed, worthless in raw ISM, born for the lane.

§5 — WHAT THE FLEET ALREADY OWNS (THE ARRAY IS NOT A NEW MACHINE)

- Phase 228's skeletal magnetosphere is the same coil hardware — re-tune the field from closed shield-bubble to open forward funnel and the ship grows its scoop out of hardware it already flies (directions §7.1).
- The Phase 249 laser is the ionizer (directions §7.2).
- The mule plasma engines are the thrower (the re-throw of the ram tax, §4).
- The array is a FIELD GEOMETRY on seated parts — no new machine is built.

§6 — BUILD SEQUENCE

- 1. FIT THE STORES at departure build: helium store plumbed to the cores' feed line; water store plumbed to the factory's electrolyzers (§7.3); Nitropril granule housings solid-state and un-pressurized (§7.4).
- 2. SIZE BY THE SIZING LAW (§2): consumption rate × time to the first gas-giant stop, plus margin — figure owed at §8 until his design pass lands.
- 3. FLY THE CLIMB on the microwave-hydrogen main engine (the Phase 01 chain); the cores are backup only and do not turn for the climb; stores drain as the ship climbs.
- 4. AT THE GAS GIANTS: re-tune the ring coils from shield-bubble to forward funnel (§7.1), bring the forward ionizer online (§7.2), collector lights — the stores stop draining and start refilling.
- 5. CRUISE: ride the seeded lane (§7.5); in-system departure/arrival eats free on the pre-ionized solar wind (§4, bonus regime).

§7 — CROSS-PHASE DIRECTIONS, INLINED (PROGRAM LAW: DIRECTIONS, NOT POINTERS)

§7.1 From PHASE 228 — THE SKELETAL MAGNETOSPHERE (the coil hardware the funnel re-tunes)

- Field coils ride the bulkhead ring frames EVERY 20 METRES, driven-mode and segmented — an artificial magnetosphere carried on the hull's skeleton; free-flying escort magnetosphere balls fly formation forward and above as a storm snowplow.

- Original ring drive: an outer ring section spinning charged mercury in rings around the ship, internal shielding, field compression toward the outer hull edge. (The Author's own caveat stands seated: mercury is inert until a hull breach — non-inductive superconductor or other material is the preferred coil path; weight and breach-safety decide.)
- FOR THIS BUILD: the same ring coils are re-tuned from the closed shield-bubble geometry to the OPEN FORWARD FUNNEL of §4. One hardware family, two field geometries.

§7.2 From PHASE 249 — THE ORBITAL ENGINE LASER (the ionizer)

- The seated laser hardware: SIDE-CENTER EXTERNAL MOUNT (the Author's correction — better than the split-gap path; the laser passes to the receptor without crossing the rotor); band receptor discipline for the oscillation circle (a free rotor precesses — the spot wanders, never pinpoints; the receptor band absorbs the wander).
- FOR THIS BUILD: the same laser family, re-tasked FORWARD of the ship as the funnel's ionizer — strip the neutral gas far enough ahead that fresh ions have room to curve into the throat (Catch One, §4). Power the ionizer as a line item — the bill is why paper ramjets die; this ship carries the turbine that pays it (phase 01 chapter).

§7.3 From PHASE 01 — THE HYDROGEN FACTORY ECONOMY (what the water store feeds)

- Electrolysers at 50 kWh per kg of hydrogen; at the 40 MW cruise rating = 800 kg/h = 19.2 t/day; at 150 MW emergency = 72 t/day. Water ledger: 9 t water per tonne of hydrogen; oxygen return 8 t per tonne. (Full directions: GP-IM-001 §3.)

§7.4 From PHASES 88 & 128 — THE NITROPRIIL DETAIL

- Phase 88: Nitropril is housed as SOLID-STATE, UN-PRESSURIZED granules — the stabilizer chemistry never rides as a pressurized gas block.
- Phase 128: Nitropril is synthesizable aboard from RAW ATMOSPHERIC NITROGEN through automated synthesizer core assemblies — no volatile liquid nitrogen tanks anywhere in the chain.

§7.5 THE SEEDED LANE (Catch Four's river)

- The lane doctrine: highway buoys and the 242 mini-moons seed the route; the ship's funnel collects from the stocked lane, not the raw ISM. Build the array for the lane; in raw ISM it is jewellery (§4, Catch Four).

§8 — REJECTED OPTIONS AND HONEST LIMITS (DO NOT BUILD / DO NOT EXPECT)

- REJECTED — a physical scoop sized to the structure: the funnel is a field, not a structure (§4).
- DO NOT EXPECT useful yield in raw interstellar medium — the density tyrant stands (grams per year). The array is born for the lane.
- DO NOT FLY the funnel past exhaust speed (50 km/s) expecting thrust — past that line the same array is a BRAKE (Catch Two).
- REJECTED — stores sized as a crutch: if the collectors do not take over at cruise, the doctrine fails honestly and the mass budget says so on day one (§2).
- HISTORY, NOT BUILD — the interregnum: this number once held the Mayernick kinetic turbine (retired by the Author's order, 13 August 2026 — 'it never ran the ship'; the old flywheel-generator description dumped the same day, preserved verbatim in sitting 246). NOTHING from the retired system is in this build. The phase number was reborn within hours — corrections never delete, but numbers are the Author's to replace.

§9 — OWED TO THE BENCH / OWED FIGURES

- The consumption rates per ship class (helium, water, Nitropril) that the sizing law (§2) multiplies — owed to his design pass.
- Throat geometry fraction (Catch Three) — the mirror-loss percentage is a design knob awaiting a chosen throat.
- Ionizer power budget at lane density (Catch One) — the laser's forward strip rate sets the funnel's usable speed envelope.

§10 — STATUS, PROVENANCE, CHANGE CONTROL

STATUS: CURRENT. Provenance: the replacement doctrine seated 13 August 2026 ('Replace phase 3 with, initial on board helium store for the plasma and water for hydrogen until speed is at cruise collection speed'); the interregnum preserved by doctrine (corrections never delete; the execution and declarations stand verbatim, sittings 246 era); the EM collection array ('the skinny') with the four catches; the propellant-split law. Nitropril family: Phases 88 & 128. Every figure and law seated from the master record, v2.1.

CHANGE CONTROL: this design changes only on the Author's orders, seated as sittings in the master record. If the master and this chapter ever disagree, THE MASTER WINS. This is a workshop chapter — it carries the current design only; the full change record lives in the master and the restart. Chapter version: v1.0, 19 August 2026.

END OF CHAPTER 03.